

Number of Clusters

The final model produces **3 clusters** covering all 8,950 customers with no noise points.

Cluster	Size	% of Customers	Business Label
0	6,797	75.9%	Everyday User
1	1,916	21.4%	Cash Advance Reliant
2	237	2.6%	Extreme / High-Value

Silhouette Scores

Metric	Value	Note
Silhouette Score	0.6899	100% data coverage
Number of Clusters	3	DBSCAN
Noise Points	0	No customers unassigned
Data Coverage	100%	All 8,950 customers
Outlier Strategy	Capping (IQR×1.5)	No rows dropped

Token Usage

Metric	Value
Total tokens	33,911
Context used	17%
Cost	\$0.00

All Rounds Progression

Round	Model	Clusters	Silhouette	Coverage	Key Technique Added
1	K-Means	3	0.2403	100%	Raw data — baseline
1	DBSCAN	5	0.1409	~89%	Raw data — baseline
2	K-Means	2	0.1894	100%	Outlier removal + Feature engineering
2	DBSCAN	12	0.2801	1.7%	Outlier removal + Feature engineering
3	K-Means	2	0.1894	100%	Hopkins + t-SNE + Davies-Bouldin
3	DBSCAN	3	0.0753	~89%	Fine-tuned eps
4	K-Means	8	0.2539	100%	Log transform + RobustScaler
4	DBSCAN	2	0.3163	~98%	Log transform + RobustScaler
5	K-Means	7	0.2697	100%	Correlation selection + UMAP
5	DBSCAN	3	0.6982	~100%	UMAP — major breakthrough
6	K-Means	8	0.2746	100%	IQR×2.0 + UMAP(nn=50)
6	DBSCAN	3	0.6562	~100%	IQR×2.0 + UMAP(nn=50)
6	HDBSCAN	3	0.6562	~100%	HDBSCAN exploration
7	DBSCAN	3	0.7369	~100%	Extended UMAP tuning
8	DBSCAN	3	0.7483	~100%	n_comp=7, min_samples=3, eps=2.0
9	DBSCAN	3	0.7483	~100%	Fine-grained eps tuning — plateau
10	DBSCAN	3	0.6899	100%	CAP instead of DROP — trustworthy